

Software Requirement Specifications

AeroVision: Drone Detection with Machine Learning



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Meeting Details

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Summary

This is the Software Requirements Specification document for AeroVision, a real-time drone detection system based on machine learning. The project goal is deceptively simple: given an image, is there a drone or not? We live in a time where drones are everywhere and people have found many ways to make use of them illegally. To this end, AeroVision trains a YOLOv8 deep learning model on one of the benchmark datasets for drone detection available on Kaggle and wraps it inside a neat Streamlit web app so that anyone can upload an image and get instant results. This document thus includes the project overview and scope, its users, competing and related systems, functional and non-functional requirements, and key use case decomposition.

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1. Introduction

Drones have become increasingly prevalent across a wide range of applications, including aerial photography, agricultural monitoring, emergency response, and military operations. However, their growing accessibility has raised serious security concerns, particularly around unauthorized drone activity near restricted areas such as military bases, airports, and private properties. Traditional security systems are not designed to detect small, fast-moving aerial objects, leaving a significant gap in surveillance capabilities. AeroVision addresses this gap by leveraging machine learning to provide an automated, image-based drone detection solution. The system accepts an uploaded image as input and determines whether a drone is present using a trained YOLOv8 deep learning model. Detection results, including bounding boxes and confidence scores, are presented through a Streamlit web interface designed to be accessible to non-technical users.

1.1. Purpose

AeroVision is designed to automate the process of drone detection in still images using a trained deep learning model, eliminating the need for continuous manual surveillance. The system addresses the limitations of human monitoring, which is prone to fatigue, inconsistency, and high operational cost. A YOLOv8 model is trained on a labeled drone image dataset to accurately identify and localize drones within uploaded images. Model performance is evaluated against a validation subset of the dataset using standard object detection metrics. The resulting system is deployed through a web-based interface that presents detection outputs in a clear, non-technical format, making it suitable for users without a machine learning background. This project also demonstrates the viability of modern convolutional neural network architectures for real-world aerial object detection tasks.

1.2. Scope

AeroVision is a desktop-based web application that allows users to upload still images and receive automated drone detection results. The system processes each uploaded image through a fine-tuned YOLOv8 model trained on a publicly available drone detection dataset sourced from Kaggle. Detection outputs include bounding boxes drawn around identified drones and a confidence score for each detection. The current version of the system is limited to single-image analysis and does not support real-time video processing. Additional features such as multi-class object detection, mobile device support, user authentication, data storage, and cloud deployment are outside the scope of this release and are reserved for future development.

1.3. Product Perspective

AeroVision is a standalone desktop application that operates independently without integration into any external system. It consists of two components: a trained YOLOv8 model stored as best.pt, which handles drone detection, and a Streamlit-based frontend that provides the user interface. The model is loaded at application startup and remains available throughout the session. Although the current version runs locally, the architecture is designed to support future integration with systems such as CCTV networks or drone management platforms.

1.4. User Characteristics

General Users (Primary Users):

- Basic knowledge of using web applications
- No technical or machine learning expertise required
- Upload images and view detection results

Security Personnel (Operational Users):

- Basic to moderate technical knowledge
- Use the system to detect drones in surveillance images
- Make decisions based on detection results for safety and monitoring purposes

1.5. Similar apps and systems/Literature Review

Several approaches to drone detection have been explored in prior research and commercial systems. In 2019, the DroneDetect dataset was introduced as one of the earliest large-scale collections of drone audio and video recordings. However, that work focused primarily on radio frequency signal analysis rather than visual object detection. DJI's AeroScope system represents a widely deployed commercial solution that identifies drones by intercepting their communication signals. While effective, AeroScope requires specialized hardware and carries significant cost, limiting its accessibility. A 2021 study by Srivastav proposed a real-time UAV detection system using YOLO-based models applied to live video feeds, demonstrating strong detection performance. However, the system required dedicated GPU hardware to meet real-time processing demands. Community-developed drone detection models are also available on Roboflow Universe, built using YOLOv5 and YOLOv8 architectures. While these models are publicly accessible, they lack a user-friendly interface and require technical knowledge to deploy. AeroVision differentiates itself by combining a trained YOLOv8 model with a browser-based Streamlit interface, providing an accessible detection tool that requires no installation or technical expertise from the end user.

1.6. Proposed Technologies

Programming Language: Python is used as the core language for the entire application.

Object Detection: Ultralytics YOLOv8 is used as the core detection model, selected for its high accuracy and minimal training overhead.

Model Training: Kaggle Notebooks provide free GPU access suitable for deep learning tasks.

Web Interface: Streamlit enables interactive Python-based applications without requiring HTML or JavaScript.

Image Processing: OpenCV handles reading images and drawing bounding boxes on detection results.

Image Format Conversion: Pillow is used to convert image formats compatible with Streamlit.

Training Dataset: A publicly available drone detection collection on Kaggle, provided in YOLO annotation format, is used for model training.

2. Requirements

The core feature of the system is the detection of drones in an image uploaded by the user. To accomplish this, the system must upload images, accept a variety of image formats, process the model for inference, and display the results clearly and simply enough for a first-time machine learning user. On the modeling side, the system must be properly trained with sufficient examples of labeled drones to learn how to generalize to unseen images. Training involves splitting the data into training and validation subsets, setting up the model, running many epochs, and saving the best-performing weights. After training and downloading the best weights, the Streamlit app loads the model once at startup and uses it for all subsequent detections, keeping response times low.

2.1. Function Requirements

The following are the functional requirements that describe what the system must do from the user's perspective.

2.1.1. Image Upload

- **Name:** FR001
- **Purpose:** The user must be able to upload an image file so the system can analyze it for drone presence.
- **User(s):** General User
- **Input:**
 - *Image File:* A JPG, JPEG, or PNG image selected from the user's device.
 - *File Size:* Must be under 10MB to ensure smooth processing.
 - *Image Format:* Only image files are accepted. Non-image files will be rejected.
- **Output:** The uploaded image is previewed on screen and passed to the detection pipeline for processing.

2.1.2. Drone Detection

- **Name:** FR002
- **Purpose:** The system must run the trained YOLOv8 model on the uploaded image and identify whether any drone is present.
- **User(s):** General User
- **Input:**
 - *Uploaded Image:* The validated image file received from the upload step.
 - *Model File (best.pt):* The trained YOLOv8 weights loaded at application startup.
 - *Confidence Threshold:* A minimum confidence value (default 0.5) below which detections are ignored.
- **Output:** The model returns bounding box coordinates and confidence scores for every drone detected in the image. If no drone is found, an empty result set is returned.

2.1.3. Result Display

- **Name:** FR003
- **Purpose:** The system must clearly show the detection results to the user in a way that is easy to understand, even without a technical background.
- **User(s):** General User
- **Input:**
 - *Annotated Image:* The output image with bounding boxes drawn around detected drones by OpenCV.
 - *Detection Data:* The list of confidence scores and drone count returned by the model.
- **Output:** The system displays the original image and the annotated output image side by side, along with the total number of drones detected and the confidence score for each one.

2.1.4. No Drone Notification

- **Name:** FR004
- **Purpose:** When the model finds no drone in the uploaded image, the system must inform the user clearly rather than showing a blank or confusing result.
- **User(s):** General User
- **Input:**
 - *Detection Result:* An empty result set returned by the model indicating zero drones were found.
- **Output:** The system displays the message "*No drones detected in this image.*" along with the original uploaded image so the user knows the analysis was completed successfully.

2.1.5. File Validation

3. **Name:** FR005
4. **Purpose:** The system must check every uploaded file before sending it to the model, to prevent errors caused by unsupported or corrupted files.
5. **User(s):** General User
6. **Input:**
 - 6.1. *Uploaded File:* Any file submitted by the user through the upload button.
 - 6.2. *File Extension:* The format of the file (e.g., .pdf, .txt, .exe, .jpg).
 - 6.3. *File Size:* The size of the file in megabytes.
7. **Output:** If the file is a valid image (JPG, JPEG, or PNG) and under 10MB, it is accepted and passed to the upload step. If the file is invalid, the system displays the error message "*Unsupported file format. Please upload a JPG, JPEG, or PNG image.*" and prompts the user to try again.

2.2 Non-Functional Requirements

Performance: The detection result must be returned within 5 seconds of the image being uploaded, assuming standard laptop hardware without a dedicated GPU.

Usability: The interface must be clean and simple enough that a user with no machine learning background can complete a detection task within 30 seconds of opening the app without any instructions.

Accuracy: The trained model must achieve a minimum mAP (mean Average Precision) of 70% on the validation set of the benchmark dataset.

Reliability: The system must handle at least 50 consecutive image uploads in a single session without crashing or slowing down.

Portability: The application must run on any machine with Python 3.9 or higher installed, on Windows, macOS, or Linux, without requiring a GPU.

Maintainability: The codebase must be organized into clearly separated modules (detect.py and streamlit_app.py) so that the detection logic can be updated independently of the user interface.

3. Use Cases and Flow of Processes

The use cases below describe the key interactions between the user and AeroVision. The primary actor in all cases is the General User, who accesses the system through the Streamlit web interface running locally on their machine. The system itself acts as a secondary actor by loading the model and processing the image.

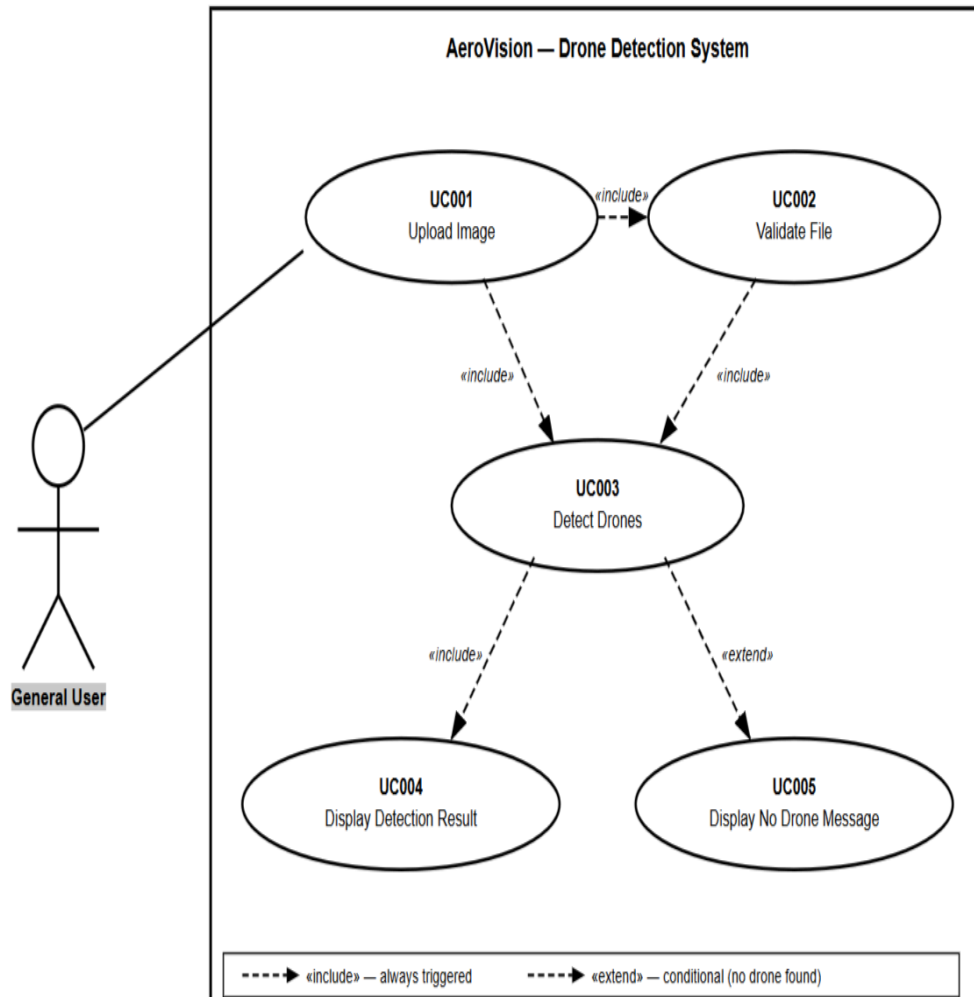


Figure 3.1: System Level Use Case Diagram

3.1 UC001 — Upload Image

ID	UC001
Name	Upload Image
Description	The user uploads an image file for drone detection through the AeroVision web interface.
Requirement(s)	FR001
Actor(s)	General User
Precondition	The AeroVision Streamlit app is running and accessible in the browser.
Postcondition	The uploaded image is previewed on screen and passed to the file validation process.
Basic Flow	1. Basic Flow 1. User opens the AeroVision app in the browser. 2. User clicks the "Upload Image" button. 3. User selects a JPG, JPEG, or PNG file from their device. 4. System previews the image on screen. 5. System passes the image to UC002 for validation. Alternative Flow : 1. User drags and drops an image directly onto the upload area. 2. System accepts it the same way as a manually selected file. Exceptions 1. User closes the file dialog without selecting anything — system stays on the upload screen with no change.

Diagram:

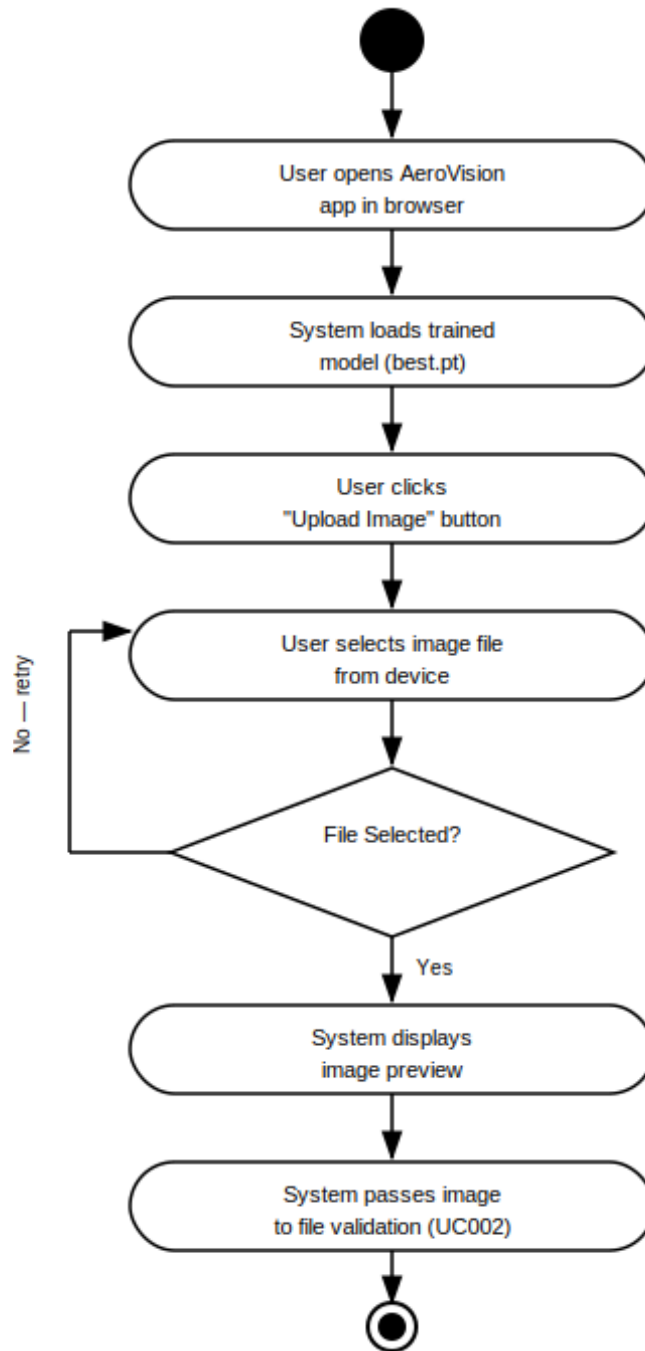


Figure 3.2 Activity Diagram UC001 : Upload Image

3.2 UC002 — Validate File

ID	UC002
Name	Validate File
Description	This use case describes how the system checks the uploaded file to make sure it is a supported image format and within the allowed size limit before passing it to the detection model.
Requirement(s)	FR005
Actor(s)	General User
Precondition	A file has been uploaded by the user through UC001.
Postcondition	If valid, the image is forwarded to UC003 for detection. If invalid, an error message is shown and the user is asked to upload again.
Basic Flow	Basic Flow 1. System reads the file extension of the uploaded file. 2. System checks that the extension is JPG, JPEG, or PNG. 3. System checks that the file size is under 10MB. 4. All checks pass — system forwards the image to UC003. Alternative Flow 1. File extension is valid but image is slightly over size limit. 2. System displays a warning and asks the user to compress or resize the image. Exceptions 1. File is not an image (e.g., PDF, TXT, EXE) — system displays: <i>"Unsupported file format. Please upload a JPG, JPEG, or PNG image."</i> 2. File is corrupted and cannot be read — system displays: <i>"Unable to read the file. Please try a different image."</i>

Diagram:

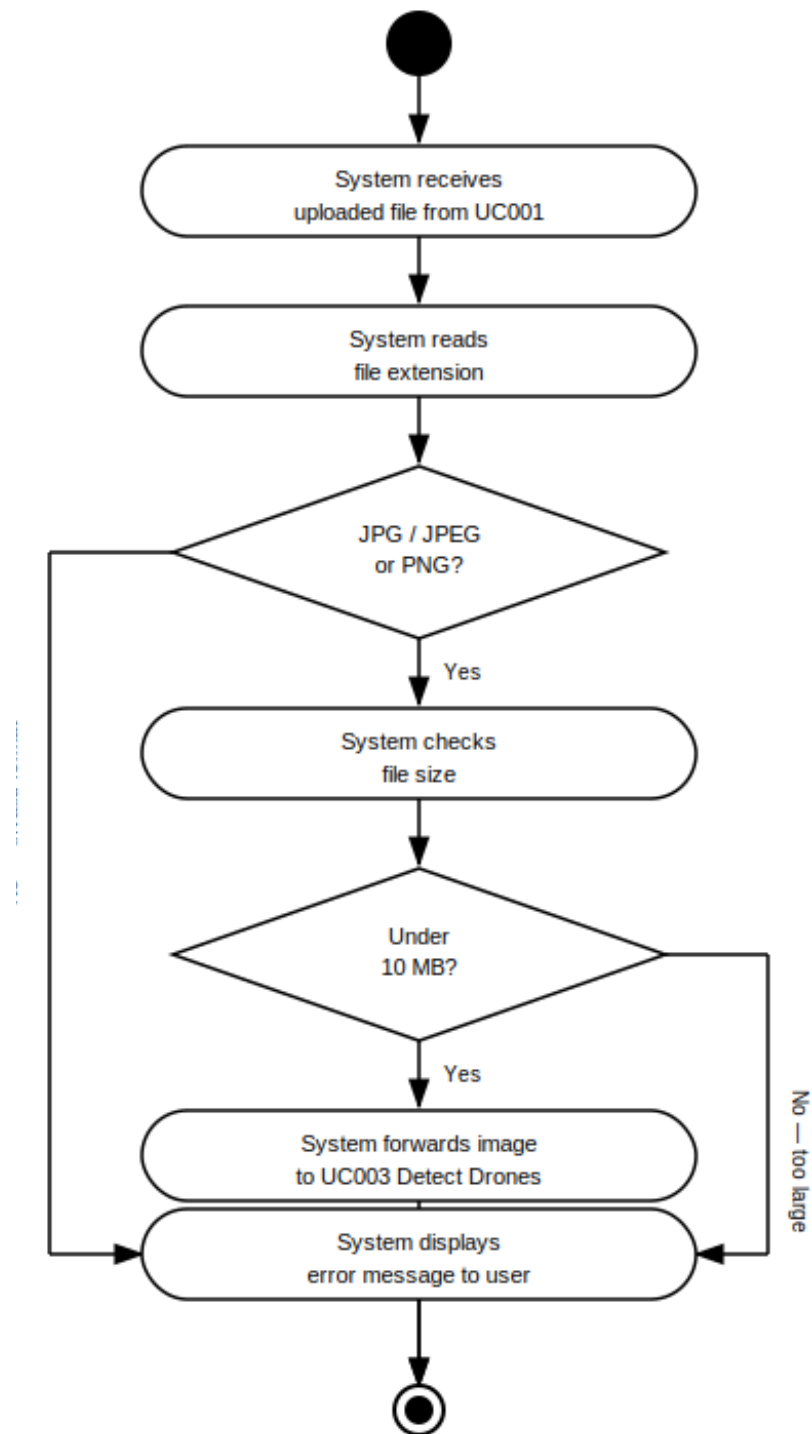


Figure 3.3 Activity Diagram UC002 : Validate File

3.3 UC003 — Detect Drones

ID	UC003
Name	Detect Drones
Description	This use case describes how the system passes the validated image through the trained YOLOv8 model to identify whether any drones are present and where they are located in the image.
Requirement(s)	FR002
Actor(s)	General User
Precondition	The image has passed file validation in UC002. The trained model file (best.pt) is loaded in memory at application startup.
Postcondition	The model returns either a list of bounding boxes with confidence scores, or an empty result. The system proceeds to either UC004 or UC005 depending on the result.
Basic Flow	<i>Basic Flow</i> 1. System receives the validated image from UC002. 2. System passes the image to detect.py. 3. detect.py loads best.pt and runs YOLOv8 inference on the image. 4. Model returns bounding box coordinates and confidence scores for all detected drones. 5. System checks if the result is non-empty. 6. System forwards the result to UC004 for display. <i>Alternative Flow</i> 1. Image is valid but very low resolution or heavily blurred. 2. Model still runs inference but returns low confidence scores. 3. System proceeds to UC004 and displays results with a note about low image quality. <i>Exceptions</i>

	1. Model file (best.pt) is missing from the model/ directory — system displays: <i>"Model not found. Please place best.pt in the model/ folder."</i> 2. Detection takes longer than 10 seconds — system displays a timeout warning.
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Diagram:

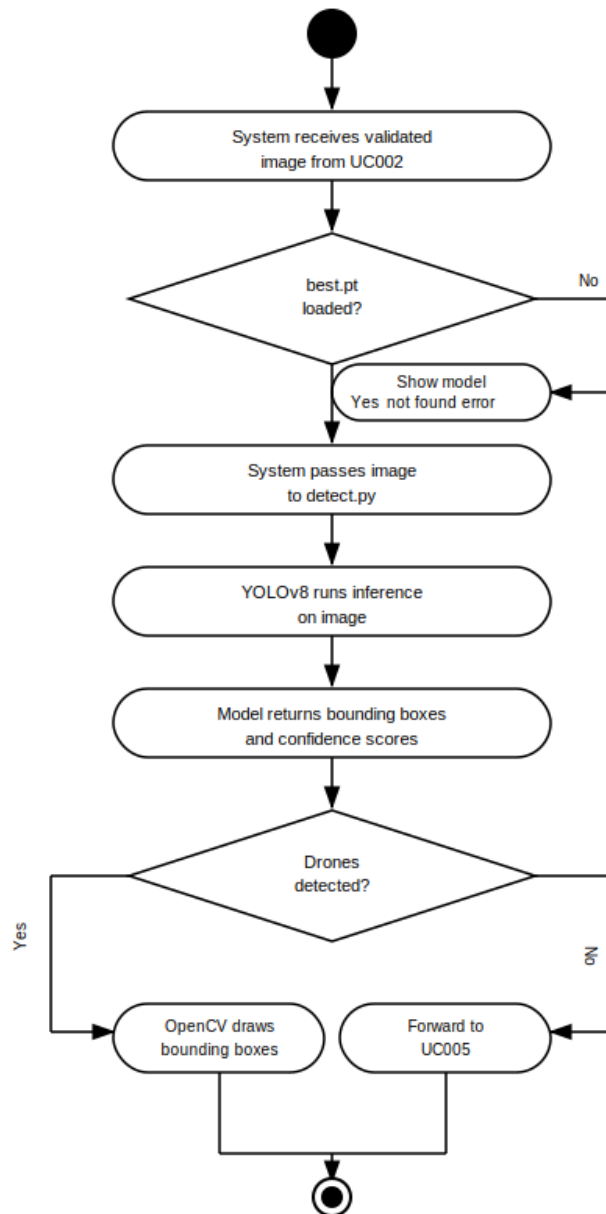


Figure 3.3 Activity Diagram UC003 : Detect Drones

3.4 UC004 — Display Detection Result

ID	UC004
Name	Display Detection Result
Description	This use case describes how the system presents the drone detection results to the user in a clear and readable format, including the annotated image and detection statistics.
Requirement(s)	FR003
Actor(s)	General User
Precondition	UC003 has completed and returned at least one drone detection with bounding box coordinates and a confidence score.
Postcondition	The user sees the annotated output image with bounding boxes drawn around detected drones, along with the total drone count and individual confidence scores.
Basic Flow	<i>Basic Flow</i> 1. System receives the bounding box coordinates and confidence scores from UC003. 2. OpenCV draws bounding boxes around each detected drone on the image. 3. Confidence score is printed as a label on each bounding box. 4. System displays the original uploaded image and the annotated result image side by side. 5. System shows the total number of drones detected and lists each confidence score below the images. 6. User reads the results. <i>Alternative Flow</i> 1. Multiple drones are detected in the same image. 2. System draws a separate bounding box for each drone and lists all confidence scores individually.

	Exceptions 1. OpenCV fails to draw bounding boxes due to an unexpected image format issue — system displays the raw detection data as text instead.
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3.5 UC005 — Display No Drone Message

ID	UC005
Name	Display No Drone Message
Description	This use case describes what happens when the detection model finds no drone in the uploaded image. Instead of leaving the user confused with an empty screen, the system clearly communicates that the analysis was completed but nothing was found.
Requirement(s)	FR004
Actor(s)	General User
Precondition	UC003 has completed and returned an empty result, meaning no drones were detected in the image.
Postcondition	The user sees a clear message confirming that no drones were found, along with the original image so they can confirm the right image was analyzed.
Basic Flow	Basic Flow 1. System receives an empty detection result from UC003. 2. System does not attempt to draw any bounding boxes. 3. System displays the original uploaded image as-is. 4. System displays the message: <i>"No drones detected in this image."</i> 5. User reads the message and understands the analysis is complete. 6. User can choose to upload a different image. Alternative Flow

	1. User is unsure whether the model missed a drone or there genuinely is none. 2. User re-uploads the same image — system runs detection again and returns the same result.
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Diagram:

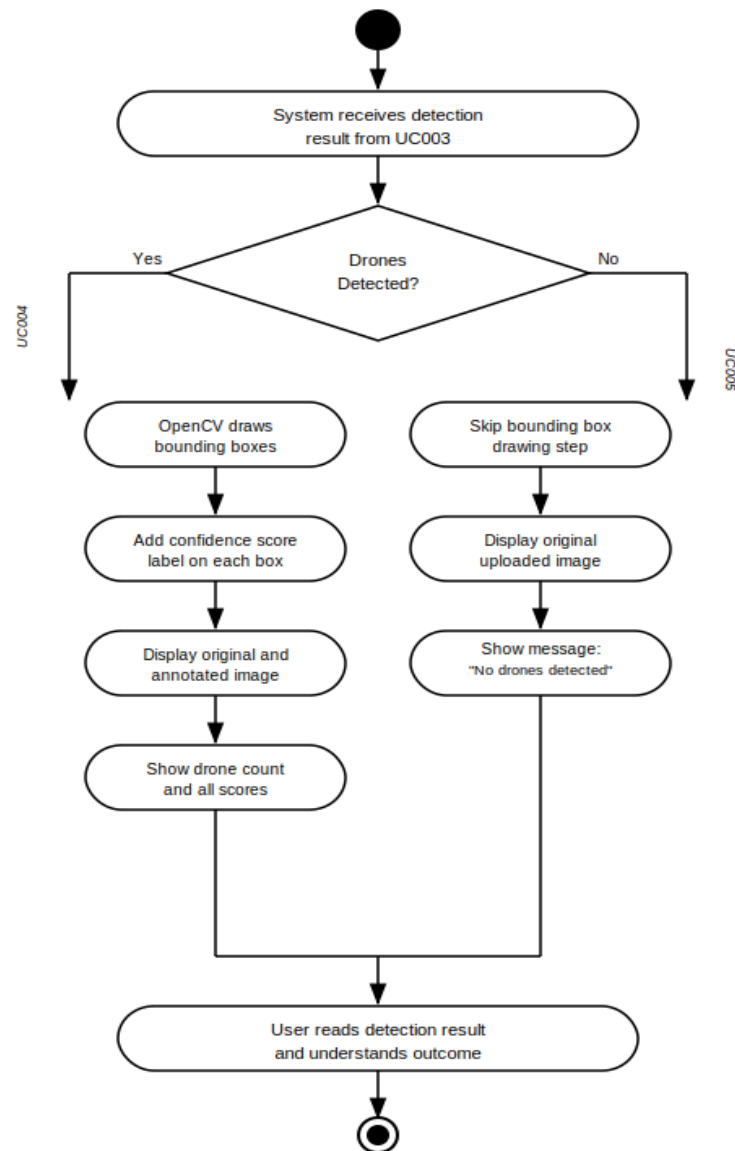


Figure 3.4 Activity Diagram UC004 & UC005 : Result Display

4. References

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